Aug 4 Union of events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Ex: How many numbers from 1 to 100 are either a multiple of 2 (or) a multiple of 3?

$A = \{1 \le k \le 100 : 2 \text{ divides } k\}$ $|A| = 50$

$B = \{ \quad " \quad : 3 \quad " \quad " \}$ $|B| = \left\lfloor \frac{100}{3} \right\rfloor = 33$

$A \cap B = \{ \quad " \quad : 6 \quad " \quad " \}$ $|A \cap B| = \left\lfloor \frac{100}{6} \right\rfloor = 16$

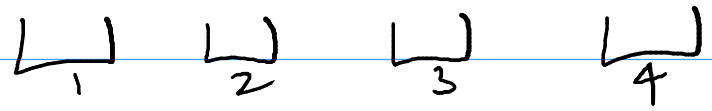
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Ex: Nos from 1 to 100 that are multiples of 2 (or) 3 (or) 5.

$A_1, A_2, \ldots, A_n$: n events

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} \sum P(A_i \cap A_j) + \sum \sum \sum_{i<k<k} P(A_i \cap A_j \cap A_k) - + \cdots (-1)^{n-1} P(A_1 \cap A_2 \cdots \cap A_n)$$

Ex: 10 balls thrown uniformly at random into 4 bins



E : no bin is empty

A_i : i-th bin is empty $\quad P(A_i) = \frac{3^{10}}{4^{10}}$

$$E = A_1^c \cap A_2^c \cap A_3^c \cap A_4^c = (A_1 \cup A_2 \cup A_3 \cup A_4)^c$$

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 4 \cdot \frac{3^{10}}{4^{10}} - 6 \cdot \frac{2^{10}}{4^{10}} + 4 \cdot \frac{1^{10}}{4^{10}}$$

Ex: n letters n addressed envelopes

letter i $\longrightarrow$ envelope i

$i = 1, \dots, n$

π : Pick permutation of $[n]$ uniformly at random

letter i $\longrightarrow$ envelope $\pi(i)$

E : n letter goes into its correct envelope

A_i = letter i goes into envelope i

$$P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$$

$$P(A_i \cap A_j) = \frac{1}{n(n-1)} \qquad i \neq j$$

$$P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \frac{1}{n(n-1)\cdots(n-k+1)}$$

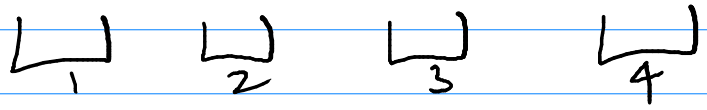
$i_1, \dots, i_k$: distinct

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = n \cdot \frac{1}{n} - \frac{1}{2} + \frac{1}{3!} - \frac{1}{4!} + \cdots + (-1)^{n-1} \frac{1}{n!}$$

$$P(E) = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - + \cdots + (-1)^n \frac{1}{n!}$$

$$\longrightarrow e^{-1} \quad \text{as} \quad n \to \infty$$

Ex: 10 balls thrown uniformly at random into 4 bins



E : exactly 1 bin is empty